

How EAE lesions *form, resolve or persist* — and what the segmentation stains add

Mouse EAE spinal cord, RRMAP2 runs 1–6 (all five runs): 67 animals, 1.38 M annotated cells, daily clinical scores, four segmentation-kit stains imaged for every section. Chronic (MOG35–55) and relapsing–remitting (PLP139–151) courses. Analyses in [notebooks/10–16](#); illustrated walk-through in [notebooks/12_lesion_story.ipynb](#).

Lesions run from an *active infiltrate* to a *fibrotic, demyelinated* state and largely resolve. Animals that do well build a *vimentin-bright astrocyte border* around their lesions, which only the images can see. Severe chronic animals keep active lesions without it, and relapse-prone animals accumulate meningeal B-cell aggregates.

- **Lesions redefined.** Against control tissue, never-immunised animals come out ~1 % lesion, down from 4–20 % with the curated labels. The lesion share falls from 0.93 at the first peak to 0.14 in long remission.
- **Six lesion states:** active monocyte-derived (S1, S4) at every peak; fibrotic / myelin-loss (S2) and glial-reactive (S0, S3) later; lymphocytic infiltrate (S5) throughout.
- **Vimentin** (*Vim* is not on the panel): lesion astrocytes carry more vimentin in milder / resolving disease in every contrast. It is the strongest correlate of chronic severity ($\rho = -0.80$). It survives batch, brightness, focus and α SMA controls, and it forms a ring at the lesion edge in milder and never-relapsing animals. The border builds between the peak and d30 only in animals that do well. Within animals, though, it sits where immune cells are active: **a response at the lesion edge, not proven containment.**
- **Chronic severity** is set at the first attack (all SEVERE peaked at 3.5) and means lesions that stayed active, not more neuropil loss.
- **Relapse** can't be predicted clinically. B cells accumulate after the first attack and move into meningeal aggregates (fewer in never-relapsing animals). Relapses are mostly new lesions, with some flaring at old borders.
- **Earlier image findings replicate** in runs 1–3: white-matter neuropil loss, astrocyte RNA-before-vimentin, T cells facing vessels, image models on unseen runs. The 18S texture marker replicates only partly.

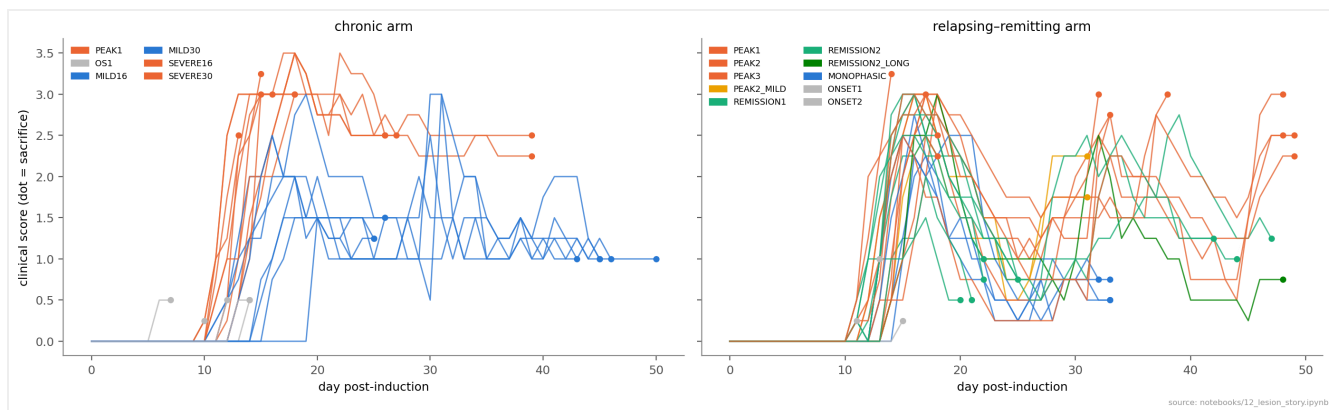


Figure 1 • Disease courses. Daily scores of every animal, coloured by sacrifice stage (dot = sacrifice). Severe chronic animals all peaked at 3.5 and never recovered; first attacks look alike in animals that will and won't relapse.

1 Lesions redefined, six lesion states

Each cell's 30-cell neighbourhood (cell-type mix, seven disease programs, cellularity) is compared with control animals of the same region (white matter, grey matter, meninges). It counts as lesion when it lies outside the control range, calibrated so held-out controls are ~1 % lesion. 72 % of cells labelled "Lesion_mix" look like normal tissue.

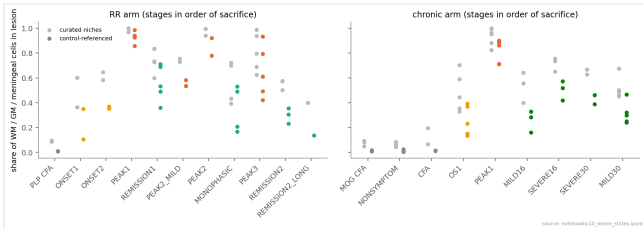


Figure 2a · Lesion share per animal, curated (grey) vs control-referenced.

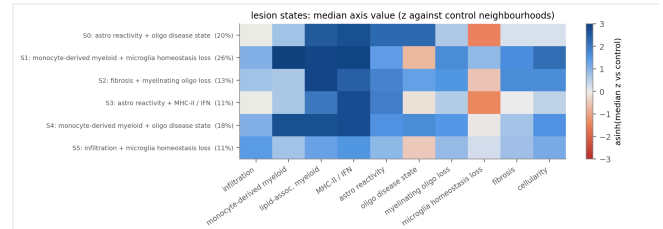


Figure 2b · The six lesion states: median axis value vs control.

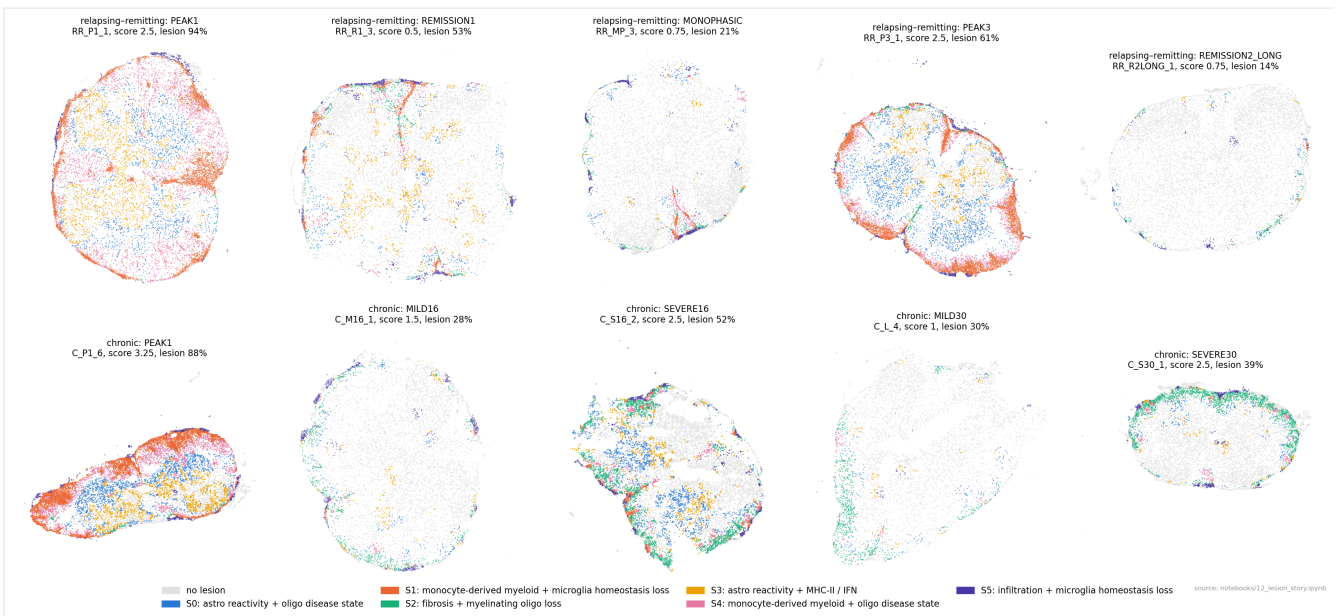


Figure 3 · Lesion states in the tissue, one representative animal per stage (median lesion share, largest piece). Active monocyte-derived lesion (orange/pink) fills the white matter at the first peak. Later, fibrotic (green) and glial-reactive (blue) tissue remains; long remission leaves small residues; chronic severe animals keep active tissue at d28–41.

2 Vimentin marks milder and resolving lesions

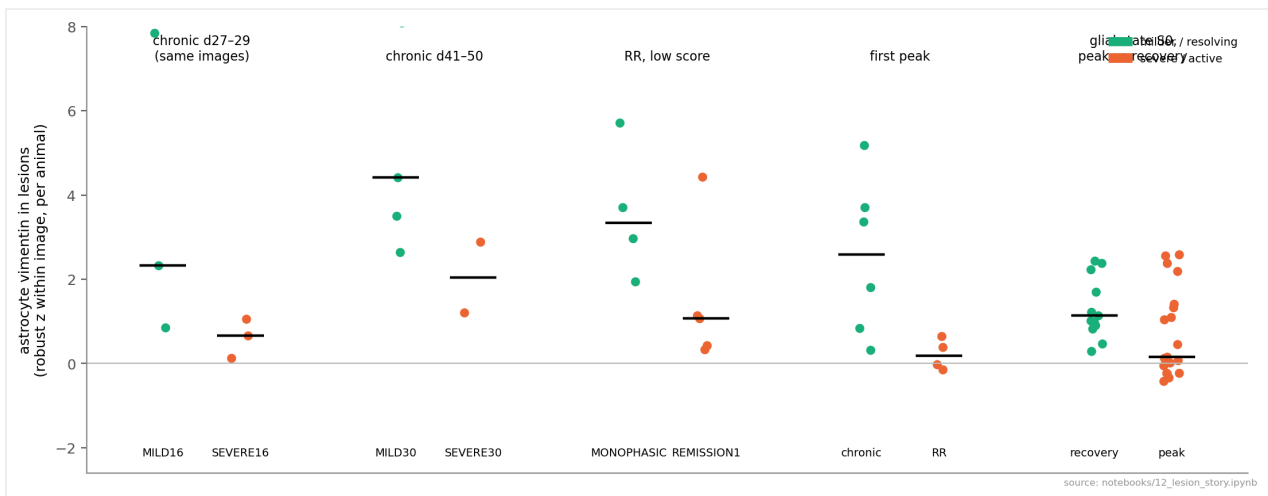


Figure 4 · Lesion astrocyte vimentin per animal in five independent contrasts; green = milder / resolving.

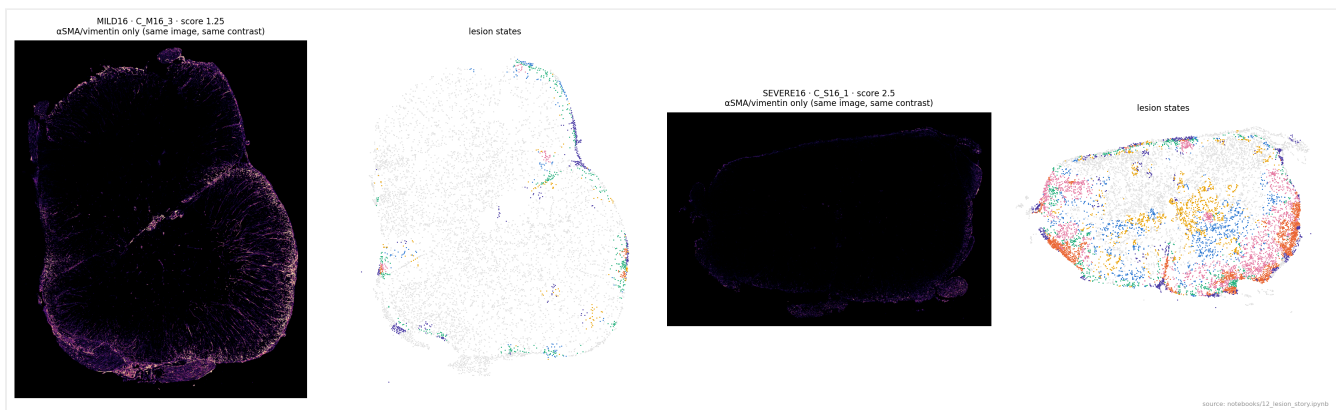


Figure 5 · Same image, same contrast: αSMA/vimentin channel of a MILD16 and a SEVERE16 piece, with lesion states. (The severe piece is dark overall; the quantitative evidence is the within-piece contrast.)

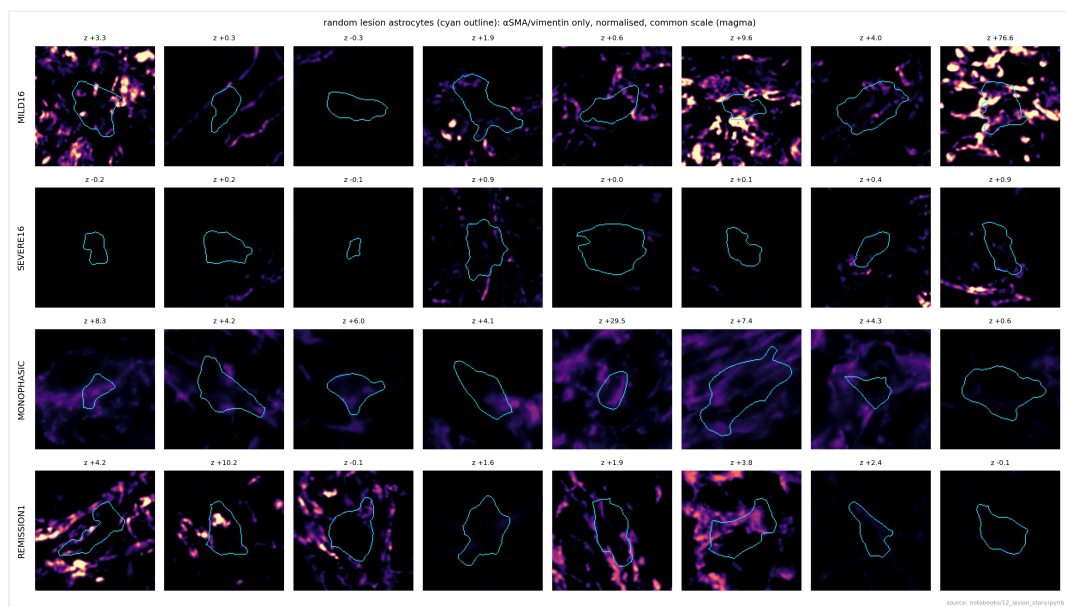


Figure 6 · Random lesion astrocytes (cyan outline), vimentin channel only, normalised per image, common scale.

Technical or α SMA? Every alternative tested (notebook 15)

The effect keeps its direction in all four contrasts within animals, far from vessels, with **no VSMC, endothelial or fibroblast cell within 20 μ m**, in deep white matter, in reactive astrocytes only, and within lesion states. Controls stay near zero: DAPI, neurons and oligodendrocytes, focus, and smooth-muscle / pericyte / fibroblast RNA in the astrocytes. A VSMC calibrator flags some channel-brightness contribution in MILD16 vs SEVERE16 and chronic vs RR; the within-animal contrast removes it.

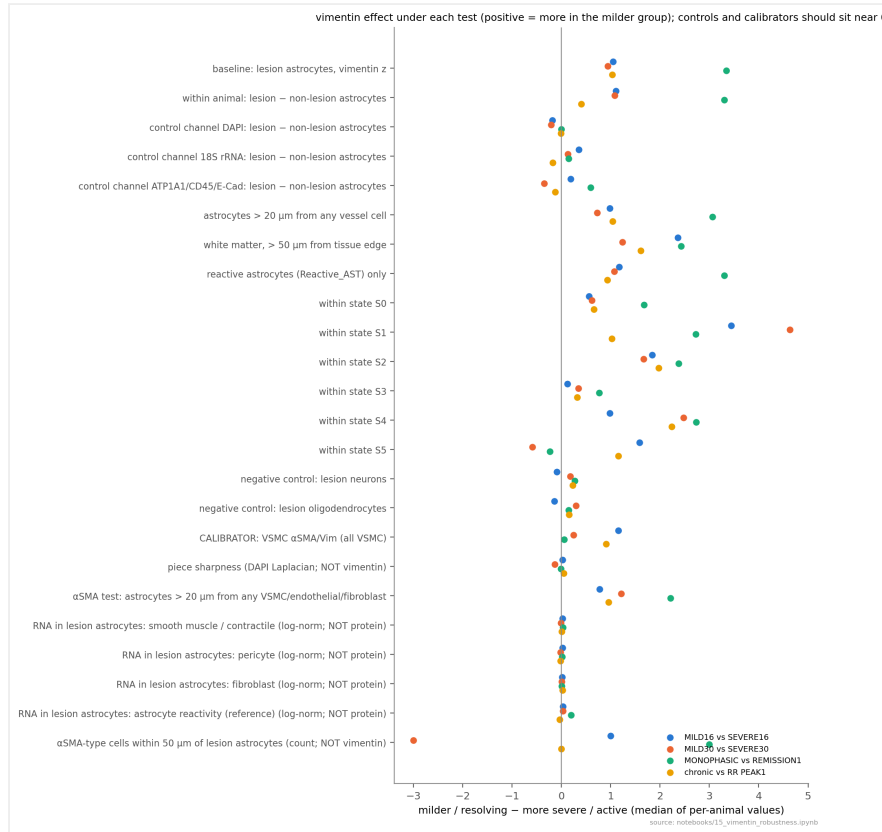


Figure 7 • Robustness. Each row a test, each colour a contrast; positive = more vimentin in the milder group.

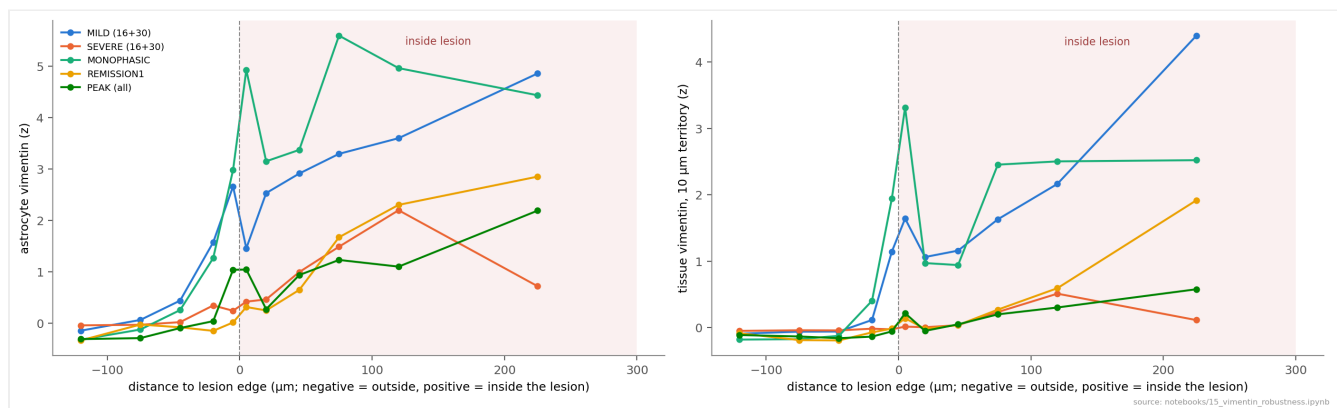


Figure 8 • A ring at the lesion edge. MILD and MONOPHASIC animals have a sharp vimentin peak in the first ~10 μ m inside the lesion edge, absent in SEVERE, REMISSION1 and PEAK animals.

3 Does the scar contain lesions? Day ~30 vs chronic peak

The border builds by d30 in animals that do well: border tissue vimentin chronic PEAK1 0.04 → MILD16 0.97 ($p = 0.02$), SEVERE16 0.02; never-relapsing MONOPHASIC 2.91 vs relapsed PEAK2 0.30 at d32–33. **But within animals**, lesions with a stronger vimentin border have *more* infiltrating immune cells just outside ($p +0.3$ to $+0.45$). This holds on deep lesions with depth as a covariate ($p +0.25$ to $+0.31$). Reading: astrocytes build the border where immune cells are active at the lesion edge. It is a response; restriction would need time-resolved data.

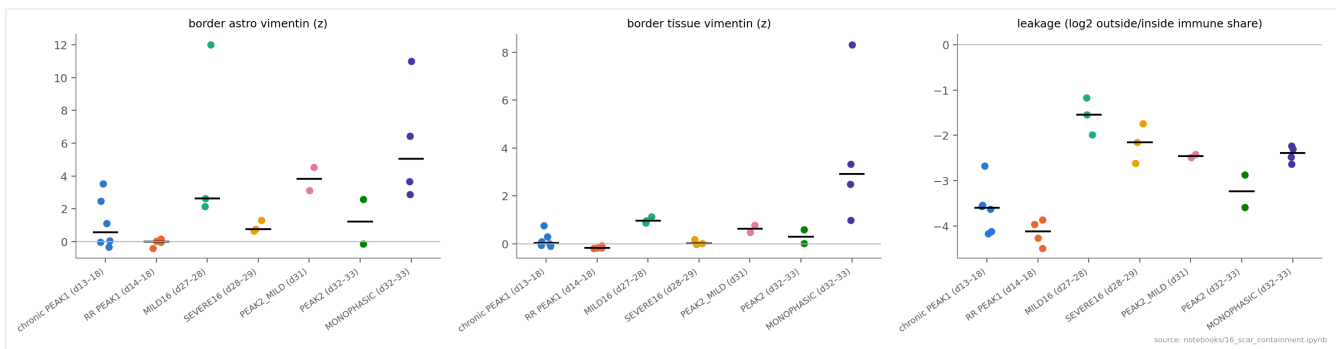


Figure 9 · Border vimentin and leakage per animal, chronic peak vs day ~30.

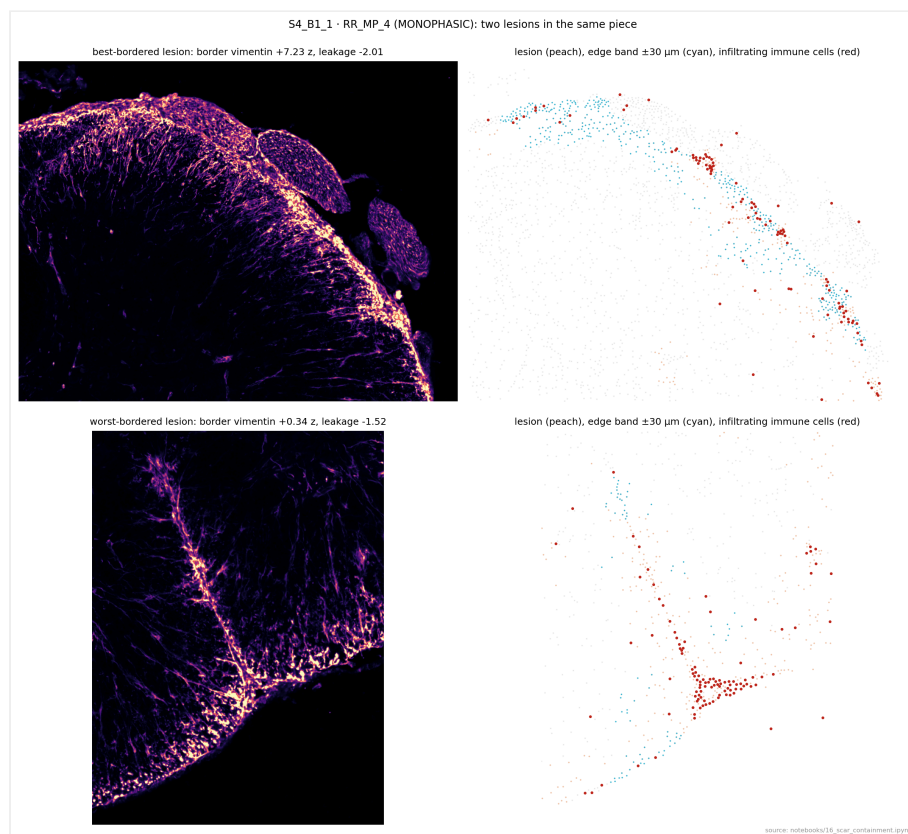


Figure 10 · Best- vs worst-bordered lesion of one MONOPHASIC animal: vimentin channel (left), lesion, edge band and infiltrating immune cells (right).

4 Chronic severity: lesions that stayed active

Severe chronic animals still carry more lesion (0.52 vs 0.28), active monocyte-derived states, T cells and dense infiltrates at d28–41, and fewer myelinating oligodendrocytes (0.08 vs 0.15 per WM cell); neuropil is not lost more. Across 13 chronic-late animals the strongest correlate of the score is **low lesion vimentin** ($p = -0.80$), then lesion share (0.58) and T cells (0.47).

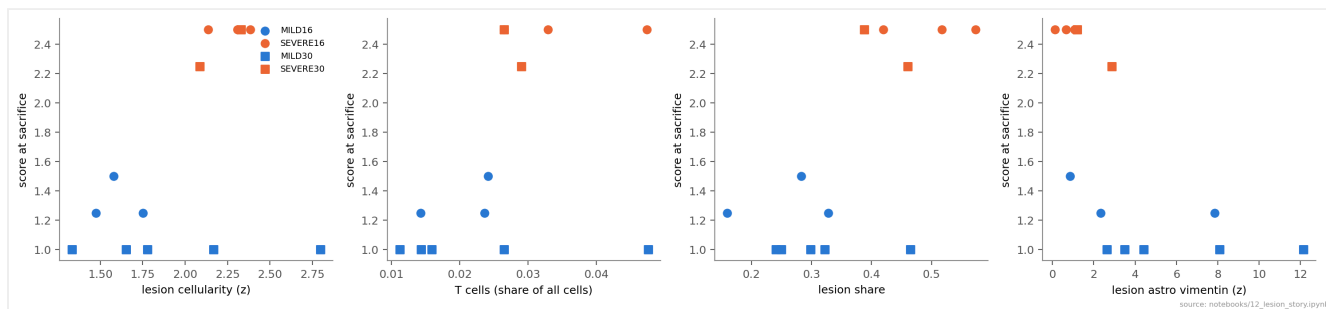


Figure 11 · Score at sacrifice vs lesion measures, chronic-late animals.

5 Relapse: B cells, and new vs reactivated lesions

B cells per 1000 cells: RR PEAK1 14 → REMISSION1 22 → PEAK2_MILD 25; aggregated share 0.26 → 0.55; meningeal share 0.16 → 0.58–0.90. Never-relapsing animals: 9 vs 22 ($p = 0.03$), deeper first recovery, more lesion vimentin. Chronic arm ≤ 8 throughout.

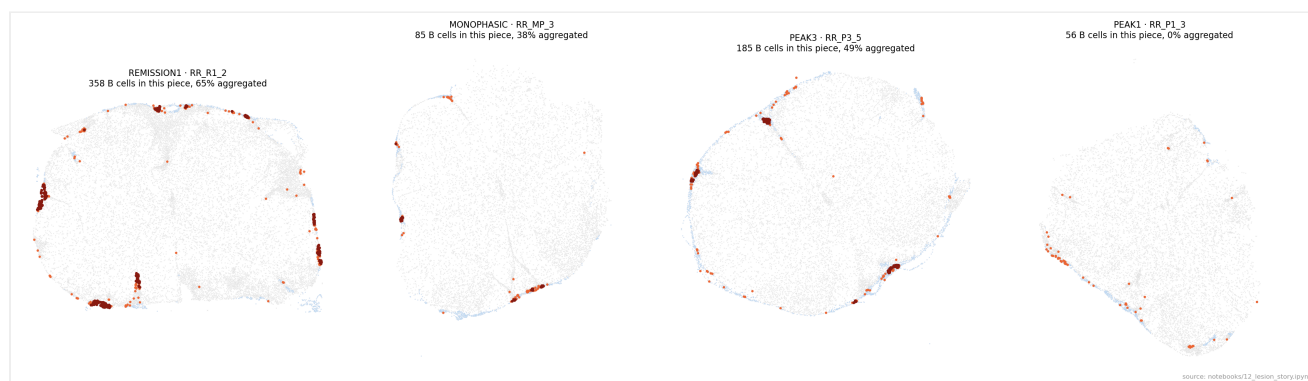


Figure 12 · B cells (red; aggregated dark red) and meninges (blue).

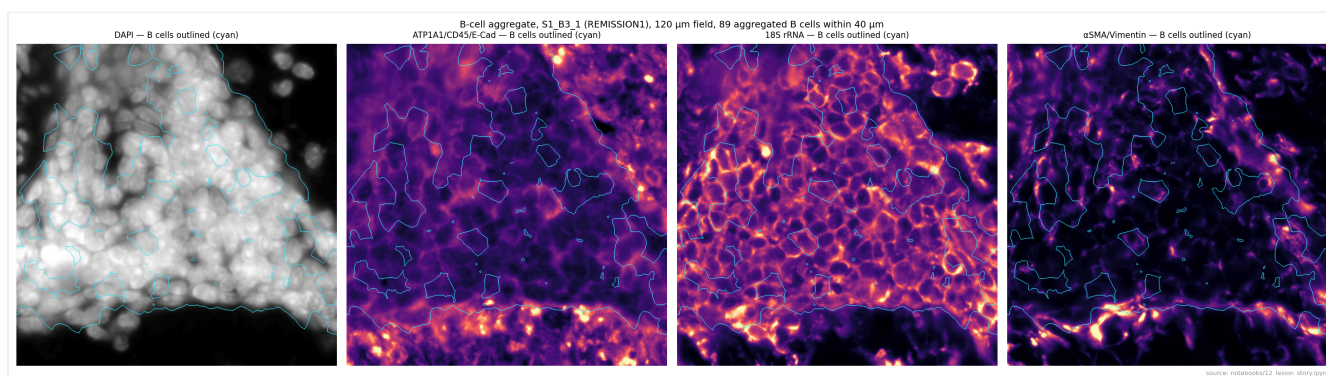


Figure 13 · A B-cell aggregate (REMISSION1), each stain separately, B cells outlined.

Relapse is mostly new lesions, plus flaring at old borders. Active tissue at relapse touches old fibrotic tissue more than at the first attack (0.23 vs 0.05 within 50 μm , $p = 4e-4$). Relapse animals have more

separate lesions (17 vs 8), and only 19 % of active relapse tissue is in lesions that contain old tissue. B-cell aggregates sit near old lesions at relapse (0.72 vs 0.31).

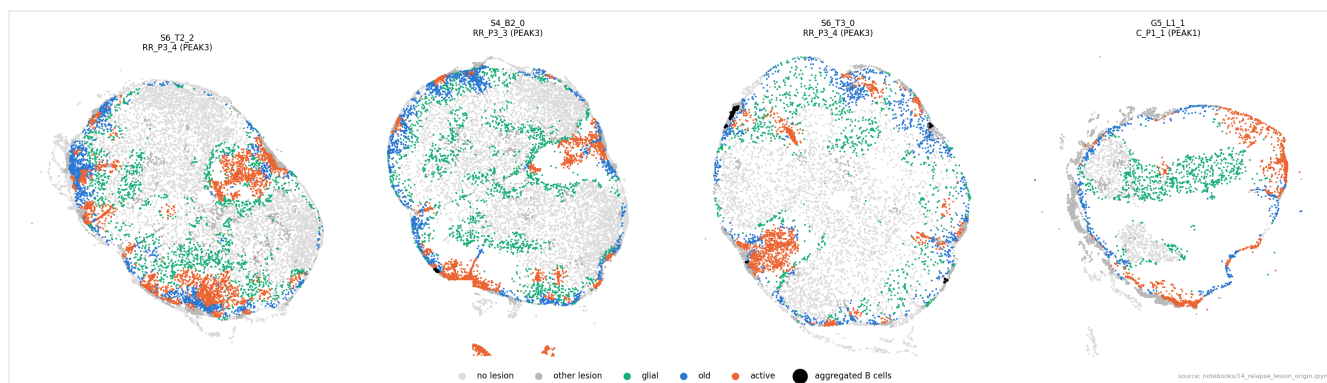


Figure 14 · Active (orange) vs old fibrotic (blue) tissue, B-cell aggregates (black): three relapse pieces and a first-attack piece.

6 Replication, damage memory, caveats

Earlier image finding	Runs 1–3
White-matter neuropil loss in lesions	replicated: $\times 0.72$ (58 pieces, $p < 1e-4$) vs $\times 0.79$
Astrocyte reactive RNA before vimentin protein	replicated: p 0.51–0.58 per run; RNA-only astrocytes more common in active disease
T cells orient 18S-rich cytoplasm towards vessels	replicated: $\cos$ 0.19 (33 animals) vs 0.14
Image-only models on an unseen run	yes, every run: lesion AUROC 0.85–0.91, cell type 0.42–0.47
18S texture severity marker	partly: astrocytes, myeloid, oligodendrocytes only

Damage memory. Within the same lesion state, neuropil trends back towards normal from peak to recovery while vimentin rises (19 vs 13 animals; best $q = 0.13$). Persisting lesions look repaired in neuropil but scarred in vimentin.

Caveats

- Small groups (2–19 animals): the evidence is consistency across independent contrasts, not single p values.
- The arms differ in antigen, likely strain, and run. MILD30 is run 1 only. MONOPHASIC and REMISSION1 never share an image.
- Lesion calls and states come from the transcriptome; the image readouts are independent of them.
- Cross-sectional design: "builds", "precedes" and "resolves" are inferred from animals sacrificed at different times.

Next

1. Validate vimentin vs α SMA with separate antibodies on a few MILD/SEVERE and MONOPHASIC/REMISSION1 sections.
2. Order events along the active $\rightarrow$ fibrotic trajectory (when neuropil, vimentin and 18S change).
3. Time-resolved or interventional test of whether the vimentin border restricts infiltration.